



Healing the Future

As technologies such as nano-tech, 3D imaging and genetic engineering become a reality, here's a projection of the future in medical technologies

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HUVRand EWI

HUVR or heads up virtual reality bridges regular visual scans with a haptic feedback device on hand for manipulation. With this medical practitioner can just look at but feel the brain in a regular MRI. Furthermore, researchers from Columbia Engineering school developed a new method to non-invasively map the electrical activation of the heart.

The technique, called Electromechanical Wave Imaging (EWI), uses ultrasound waves to detect very small deformations following the electrical activation of the heart. It can be incorporated into most current ultrasound scanners. Previously these measurements were possible only through invasive electrode contact.



Stemcells

Stem cells are primordial cells that divide and turn into other organs. Stem cells have the potential to be turned into any tissue in the body, an ability that has led researchers to believe they could be used to make "spare parts". Along with progress in genetic engineering, the idea of designer babies could eventually become a reality

Lipta woke up with a headache, feeling very uneasy. Thought she didn't know what it was yet, she knew something was wrong. She went to her trusty PDD (Personal Diagnostic Device) which measured her temperature, heart rate, respiration and some other parameters she wasn't even aware of. The measurements were saved on a remote database, and since her heart rate was unusual, her whole medical record was automatically forwarded to her doctor- Dr. Shah

She set up a video conference with her doctor, who prescribed some medicines available from vending machines, but called her for a check-up later in the day. At her appointment, he studied a 3D visual of her brain scans from the MRI using HUVR. Also, through laser endoscopy, 3D images of the nether organs were obtained and checked for defects. After that, through EWI, her heart was checked for minute deformations.

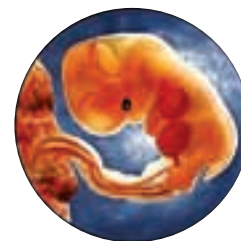
Although it was found later that Lipta was fine, she lived in an age of complex viruses and even more complex cures. Due to easy movements of humans, climate conditions, closer proximity to animal habitats, and evolutions of viruses to develop immunity for certain cures, diseases evolved and spread rapidly. As new cures were found, newer diseases were discovered.

Cures for cancer and AIDS were developed with the help of cloud computing projects like caBIG (Cancer Biomedics Informative Grid). Nanotech devices were released inside the body which could kill of unwanted cancer cells as soon as they grew.

A couple of days later, Lipta was back again, this time with her friend Kavya who was to undergo an operation. Her surgery right from anaesthetising the patient to making incisions was performed by a surgeon sitting in Denmark via human controlled robots. Robotics had come a long way since medical robots like nuvoSurge, Da Vinci and McSleepy were first used in surgery. The surgery was successful and as always 100 per cent accurate.

"Too bad", the doctors discussed amongst themselves, that "people like Lipta and Kavya are not like the transhumans of today who were the benefactors of the progress made in genetic engineering and cyborg search". Today, designer babies were conceived based on the selection of desirable genes, along with customization in gender, skin and colour etc. Movies such as *Gattaca* that predicted such a day were no longer a crazy man's imagination. Genetic engineering as always was subject to a lot of derision and debate, but its advantages were obvious. But still, scientists were able to clone parts of the human body and through stem cell growth, could find viable cures for conditions as diverse as Parkinson's disease, diabetes and spinal cord injury [<http://bit.ly/digitstem>]

As the definition of human beings expands, medicine will continue to evolve. As new cures and diseases keep being discovered, one trend remains constant, that man has been able to predict his future by actually creating it.



Designer babies

This technology is still far away but basically it implies that some time in the future you could potentially choose the most desirable genes for your offspring thereby increasing resistance, selecting certain traits and much more. One of the small victories in this field comes in the form of a break-through that could have huge implications. British and Canadian scientists have found a way of reprogramming skin cells taken from adults, effectively winding the clock back on the cells until they were in embryonic form.



Robot Surgery

A team of surgeons at McGill University, including the da Vinci robo-surgeon and a robot anaesthesiologist named, McSleepy, recently removed a person's prostate during what is being billed as the world's first all-robotic surgery. The device transmits hi-def 3D images to nearby workstation, where it is controlled by surgeons with a precision that cannot be provided by humans alone.